

Most exploded on contact with a ship, but some were detonated via insulated wires controlled by agents hidden onshore. Torpedoes sank or damaged 43 Union vessels in all, making them the Confederacy's most effective defense against Union naval superiority. Torpedoes were also used in direct attacks. Fixed to the end of a long underwater spar that protruded forward from the bows of specially designed boats, the torpedo was rammed into the enemy ship. In 1863, General P. G. T. Beauregard, commander of Charleston's defenses, hoped to use a semi-submersible torpedo boat, CSS *David*, to break the Union stranglehold on the harbor. On the night of October 5, the *David* attacked the Goliath of the blockading squadrons, USS *New Ironsides*, exploding 100lbs (45kg) of gunpowder against her starboard side. The damage was minor, but more *David*-style craft, forerunners of modern torpedo boats, followed.

Spar torpedoes were used by both sides. On the night of October 27, 1864, Lieutenant William B. Cushing and 14 volunteers from the U.S. Navy entered the Roanoke River on a boat fitted with a spar torpedo. Their target was the CSS *Albemarle*, a Confederate ironclad that had started driving off the Union naval forces on the North Carolina shore. With only a 6-ft (1.8-m) draft, the *Albemarle* was able to navigate rivers where no Union warship could follow. At 3 a.m., braving small-arms fire and the ironclad's roaring cannon, Cushing sank the monster with his well-placed torpedo.